The algebraic Robbins Hankel conjecture

Principia Math In Progress

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MathDB 350292. A proof of Barry's Conjecture 32.

Task 1: the full determinant identity

For a formal power series D with $D(0) = 1$, define its revert transform by

$$\mathcal{R}(D)(x) = \frac{(tD(t))^{\langle -1\rangle}(x)}{x}, \qquad H_n(D) = \det([x^{j+k}]D)_{0\le j,k\le n}.$$

The inverse is compositional. Set $H_{-1}(D) = 1$, and let

$$A_n = \prod_{k=0}^{n-1} \frac{(3k+1)!}{(n+k)!}, \qquad A_0 = 1$$

be the Robbins numbers. All series below are formal over $\mathbb{Q}$, extended to $\mathbb{Q}(i)$ when needed.

Use the constant-term-one square root to define

$$F(t) = \frac{\sqrt{1+2t+5t^2}-1-t}{2t^2}, \qquad A(t) = \frac{1}{1-t^2/(1+t^2F(t))}. \tag{1}$$

Then $F(0) = 1$ and

$$t^2F(t)^2 + (1+t)F(t) - 1 = 0. \tag{2}$$

Consequently $F = 1/(1+t+t^2F)$, so $A(ix)$ is precisely the normalized continued fraction in [1, Conjecture 32]:

$$A(ix) = \cfrac{1}{1+\cfrac{x^2}{1-\cfrac{x^2}{1+ix-\cfrac{x^2}{1+ix-\cdots}}}}.$$

Theorem 1. *For every integer $n \ge 0$,*

$$\boxed{H_n\big(\mathcal{R}(A(i\,\cdot))\big) = A_n.} \tag{3}$$

Equivalently, if $U = \mathcal{R}(A)$, then $H_n(U) = (-1)^{n(n+1)/2}A_n$.

Reduction to a rational reversion

Let

$$G(t) = \frac{1+t}{1+t-2t^2-t^3}, \qquad V = \mathcal{R}(G).$$

We prove the exact identity

$$U(x) = \frac{1}{1+x^2V(x)}. \tag{4}$$

Indeed, put $u = xU(x)$, so that $uA(u) = x$. From (1),

$$U = 1 - \frac{u^2}{1+u^2F(u)}.$$

Since $U = 1 - x^2 + O(x^3)$, the series $V_0 = (U^{-1}-1)/x^2$ has constant term one. The last equation gives $V_0 = U/(1+u^2F(u))$, hence

$$F(u) = \frac{U/V_0 - 1}{u^2}, \qquad u = \frac{x}{1+x^2V_0}.$$

Substitution in (2), followed by clearing denominators, gives

$$x^3V_0^3 + 2x^2V_0^2 + xV_0^2 - xV_0 + V_0 - 1 = 0,$$

or

$$(1-V_0)(1+xV_0) - x^2V_0^2(2+xV_0) = 0. \tag{5}$$

Temporary division by x^2 is legitimate in the Laurent series field; all resulting expressions are the stated power series. Equation (5) is equivalent to $xV_0G(xV_0) = x$. The unique compositional inverse with linear coefficient one therefore gives $V_0 = V$, proving (4).

Deleting one Hankel row and column

For any series D, its Hankel matrix has generating kernel

$$K_D(x,y) = \frac{xD(x) - yD(y)}{x-y} = \sum_{j,k\geq 0} [t^{j+k}]D(t)\, x^j y^k.$$

Identity (4) implies

$$\frac{K_U(x,y)}{U(x)U(y)} = 1 - xyK_V(x,y). \tag{6}$$

Multiplication of a matrix kernel by $1/U(x)$ on the left and $1/U(y)$ on the right is congruence by the lower triangular Toeplitz matrix of $1/U$. Its diagonal entries are one. For each finite leading principal block, (6) therefore yields

$$H_n(U) = (-1)^n H_{n-1}(V). \tag{7}$$

This also holds at $n = 0$ with the empty determinant convention.

The established rational input and the signs

Barry's later paper [2, Proposition 9] proves the known formula

$$J(t) = \frac{1-t}{1-2t-t^2+t^3} \quad \Longrightarrow \quad H_m(\mathcal{R}(J)) = (-1)^{m(m+1)/2} A_{m+1}. \tag{8}$$

We explicitly credit this rational-series evaluation. Its application to G can also be read in [2, Proposition 14]. For completeness the required invariance is proved here. Put $B(t) = G(-t)$ and $W = \mathcal{R}(B)$. Direct algebra gives $J = B/(1-tB)$. Inverting the functions $tJ(t)$ and $tB(t)$ consequently gives

$$\mathcal{R}(J)(x) = \frac{1}{1+x} W\left(\frac{x}{1+x}\right). \tag{9}$$

Its coefficient of x^k is $\sum_{j=0}^{k} \binom{k}{j}(-1)^{k-j}[x^j]W$. The resulting Hankel matrix is the congruence of the Hankel matrix of W by the unipotent Pascal matrix $P_{kj} = \binom{k}{j}(-1)^{k-j}$. Thus these Hankel determinants agree. Also $V(x) = W(-x)$; scaling coefficient k by $(-1)^k$ multiplies a size $m+1$ Hankel determinant by $(-1)^{m(m+1)} = 1$. We conclude from (8) that

$$H_m(V) = (-1)^{m(m+1)/2} A_{m+1}.$$

Combining this with (7) proves $H_n(U) = (-1)^{n(n+1)/2} A_n$. Finally, compositional inversion commutes with coefficient scaling: $\mathcal{R}(A(c\,\cdot))(x) = U(cx)$ for $c \neq 0$. At $c = i$, the determinant multiplier is $i^{n(n+1)} = (-1)^{n(n+1)/2}$. The two signs cancel, giving (3) at every $n \geq 0$.

The first real coefficients are $1, 0, -1, 0, 3, -1, -12, 11, 51, -89, -204, 628, 646$, as in [1]. The proof uses the normalized definition (1) and the exact Conjecture 32 continued fraction. The stray numerator x in the preceding real continued-fraction display in [1] is not used as grounds for any counterexample or research credit.

Tasks 2–5: proposed mathematical follow-ups

Task 2: Classify the parameters in quadratic tails $at^2F^2 + (1+bt)F - 1 = 0$ and transforms $A = 1/(1 - ct^2/(1+dt^2F))$ for which the reduced series $(\mathcal{R}(A)^{-1} - 1)/x^2$ is the reversion of a rational function.

Task 3: Seek a weighted extension of (3) whose determinants enumerate alternating sign matrices by their number of negative entries, and determine the corresponding tail equation.

Task 4: Determine explicit three-term recurrence coefficients for the formal orthogonal polynomials associated with U, including the coefficients not determined by the Hankel products.

Task 5: Construct a direct cancellation or nonintersecting-path proof of (3) from the continued fraction, explaining why the surviving objects are counted by A_n.

References

[1] P. Barry, *Centered polygon numbers, heptagons and nonagons, and the Robbins numbers*, arXiv:2104.01644v2 (7 April 2021), Section 10, pp. 40–41, Conjecture 32.

[2] P. Barry, *From Fibonacci to Robbins: Series Reversion and Hankel Transforms*, Journal of Integer Sequences **24** (2021), Article 21.10.2, Propositions 9 and 14, pp. 8–11. This is the shortened and updated publication of [1]; the algebraic-series Conjecture 32 is not carried over.